

Convex Core, from the word up

What “convex” means, how it lives inside $w_{\text{equity}} + w_{\text{convexity}} + w_{\text{duration}} = 1$, and why these specific ETFs.

First — the problem we’re actually solving

Before the word and the equation, it’s worth being honest about why a long-term investor needs *anything* beyond “buy good things and hold.” Three problems push back on the simple answer.

1. Buy-and-hold wins on the chart, but you live on the *path*

Over a long enough horizon, broad equities compound and buy-and-hold “wins” on paper. But you don’t get to live in the average — you live through the **path**. The S&P fell about **–55%** in 2008–09 and roughly **–50%** in the dot-com bust, and the arithmetic of digging out is brutal: a 50% loss needs a **+100%** gain just to get back to even. A long enough horizon survives that. The trouble is that most real money has a *shorter* effective horizon than the chart implies — because of the next two problems.

2. Sequence-of-returns risk — the *order* matters, not just the average

Two investors can earn the *exact same average return* and end up in completely different places if the **order** of those returns differs — especially once money is being withdrawn (retirement) or spent unevenly. A large loss **early** does far more damage than the same loss late, because it shrinks the base that everything afterward compounds on. Buy-and-hold has no defense against a bad sequence; it simply takes whatever order the market hands it. A –55% in the first year of drawing down is a different life than a –55% in year twenty.

The first two together: the goal isn’t just a high average return — it’s **protecting the path**, because deep drawdowns and bad sequences quietly do the real damage.

3. You can’t fix it by *predicting* — almost every fitted strategy fails

The obvious instinct is “just get out before the crash.” But strategies that try to **forecast** — by fitting parameters to history, optimizing on the backtest, or training a model to find “the signal” —

almost all fall apart out of sample. Markets are **non-stationary** (the rules keep changing), the signal is weak, and tuning a model to the past is mostly just **overfitting** it. This isn't a hunch: we tested it directly and ruled it out — machine-learning rankers, volatility forecasters, trend-clarity features, multi-strategy blends — **five pre-registered attempts, every one failed its honest baseline**. The lesson is not “find a better crystal ball.” It's that **prediction is the wrong tool**.

So what's left? If you can't reliably predict the crash, and you can't afford to simply ride it, the move is to change the **shape** of your returns — make the crashes hurt less *without forecasting anything*. That shape has a name, and it's the entire idea behind this model: **convexity**.

1 · Start with the word: *convex*

Convex just means **curved** — and in investing it means a payoff that curves **in your favor**: you give up only a little when you're wrong, and the protection grows fastest exactly when you need it most. The opposite of a straight line.

“Own the market” (buy-and-hold) is a **straight line**: if the market falls 37%, you fall 37%. Convex Core deliberately **bends that line** so a 37% market crash costs you a fraction of that — while you keep most of the upside. That bend is the whole idea.

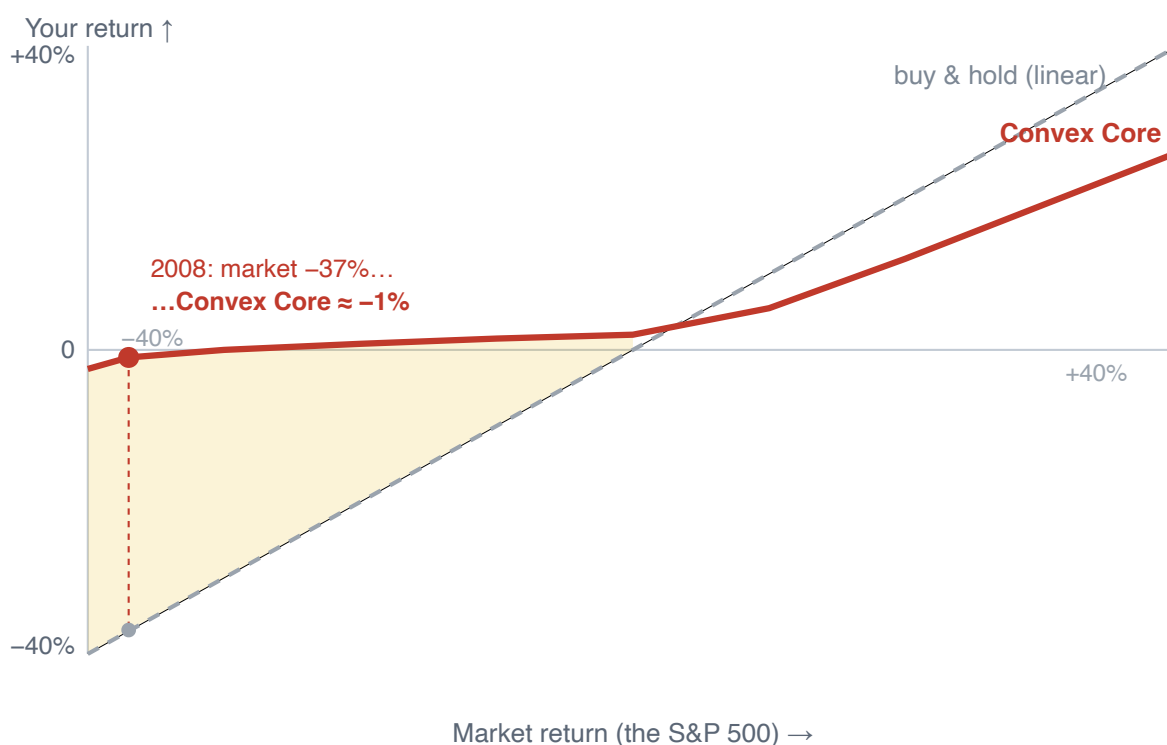


Figure 1. The straight grey line is buy-and-hold. The red curve is Convex Core. On the **downside (left)** the curve lifts away from the line — that gold gap is the **convexity**: crash protection that grows as the crash deepens. On the upside it gives up a little. (Illustrative shape; the 2008 point $-37\% \rightarrow \approx -1\%$ is the real backtest.)

The intuition in one line: a straight line treats a -37% crash and a -5% dip the same way (proportionally). A **convex** payoff fights *hardest* in the deep left tail — exactly where buy-and-hold hurts most.

Why the tail is worth obsessing over

Losses are **asymmetric**: the gain you need to recover grows much faster than the loss. Lose 20% and you need +25% back. Lose 50% and you need **+100%**. Avoiding the deep holes matters more for long-run wealth than catching every last rally.



Figure 2. Recovery math is convex too. A buy-and-hold investor down 55% (SPY, 2008–09) needs +122% just to break even; Convex Core's worst drawdown of $\sim -18\%$ needs only +22%.

2 · How that bend lives in the equation

$$w_{\text{equity}} + w_{\text{convexity}} + w_{\text{duration}} = 1$$

This is a **budget**. Every dollar is fully invested (the weights sum to 1 — Convex Core is long-only, no leverage), and it splits across **three jobs**:

Term	Its job	Shape
w_equity	Participate. Own the market so you compound with it (~73% of years are up).	Linear — this is the straight line.
w_convexity	Profit from chaos. Trend-following that goes short falling assets — it <i>pays</i> in sustained crashes.	Convex — bends up in big moves.
w_duration	Catch the flight-to-quality. Treasuries that rally when scared money flees stocks.	Convex (conditionally) — the deflation hedge.

So the bend in Figure 1 isn't magic — it's **two of the three terms**. The equity term is the straight line; **w_convexity + w_duration** are the hedges that bend it upward in the tail.

The canonical split (the model's a-priori default)

equity 0.65

cvx 0.15

duration 0.20

Notice **w_duration** is the **plug**: the model sets **w_equity** and **w_convexity**, and duration is simply $1 - w_equity - w_convexity$ — whatever budget is left after participation and crisis-alpha. That's why the equation is written the way it is: it's a constraint, not three free choices.

Two dials sit on top of this static split:

- ① **The glidepath** moves **w_equity** for your life-stage — 0.65 (balanced) up to **0.95** (accumulation, the live IRA setting) and back down toward 0.65 as you near drawdown years.
- ② **The volatility brake** scales the equity exposure *down* automatically as markets get turbulent:

$$eq_scale = \min(1, vol_target \div realized_vol)$$
with $vol_target = 12\%/yr$.

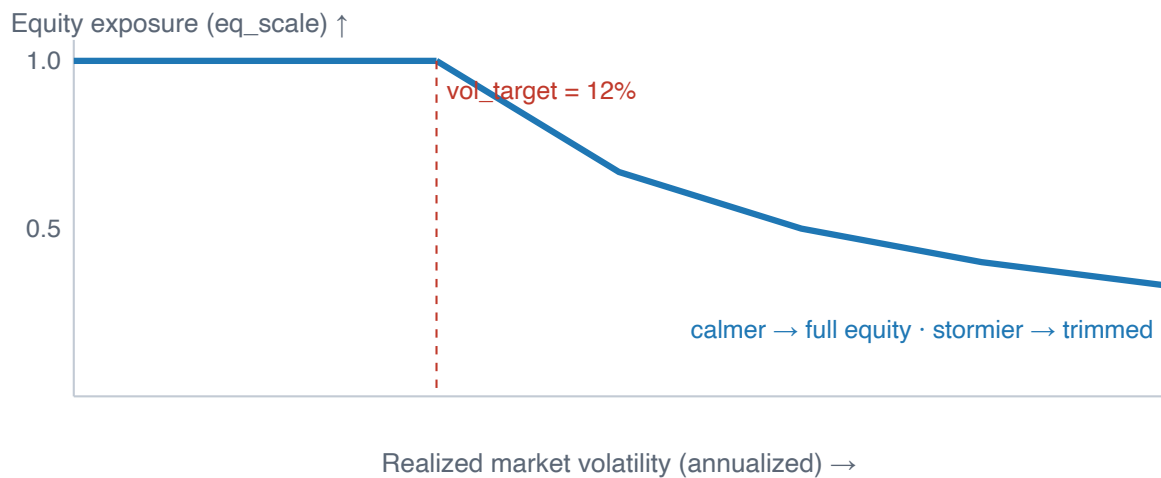


Figure 3. The brake. While realized volatility is at or below the 12% target the portfolio holds full equity; once volatility spikes, equity is scaled back proportionally — in *days*, with no forecast required. (A confirmatory hard cap of 0.70 applies in liquidity-stress regimes.)

3 · How the specific ETFs fit each term

Each term is filled by instruments chosen for the *shape* they contribute — not for a return forecast.

Term	ETFs	Why it has that shape
equity 0.65	SPY core (60% of the sleeve) + up to 5 IC-scored factor/sector tilts (40%) — currently VLUE · XLK · MTUM · PDBC · USO , with a guaranteed value slot.	SPY is the straight line — broad market participation. The tilts lean toward factors the panel measures as currently rewarded; a de-dup guard stops near-twins (e.g. it keeps no-K-1 PDBC over DBC).
convexity 0.15	DBMF (50%) · KMLM (30%) · BTAL (20%). Before these funds existed, the model uses a point-in-time proxy basket TLT / GLD / DBC (what trend-followers actually hold).	DBMF & KMLM are managed-futures / trend-following — they go <i>short</i> assets that are falling, so a sustained crash is their best environment. That is a literally convex payoff: a small bleed in calm chop, a big positive when a trend (up <i>or</i> down) persists. BTAL is long-low-beta / short-high-beta — it rises when the market falls.
duration 0.20	Regime-switched: neutral → IEF 60 / BIL 40 · risk-off rally	Treasuries are the classic flight-to-quality asset — they rally when investors flee stocks (2008,

→ TLT 50 / IEF 40 / BIL 10 ·
inflation → VTIP 50 / BIL 30 /
 IEF 20.

2020). The model lengthens duration (TLT) when that hedge is working and switches to **TIPS (VTIP)** when inflation would make nominal bonds *fail* (2022).



Figure 4. The convexity sleeve on its own. Trend-following has a **smile**: it pays in big sustained moves (its best case is a crash) and costs a small premium in flat, directionless markets — the textbook shape of *positive convexity*, like owning a straddle.

4 · Why *two* hedges, not one

The single most important design decision: **no one hedge works in every crisis**. Bonds are a brilliant hedge — until inflation, when they crash *with* stocks (2022). Trend-following is a brilliant hedge — but needs a *sustained* move to work. Hold both, and at least one is firing in almost every kind of selloff.

Crisis type	Equity	Convexity (trend)	Duration (bonds)
2008 — deflationary deleveraging	crashed	helped	helped a lot
2020 — COVID liquidity shock	crashed	mixed	helped
2022 — inflation / rate shock	fell	helped a lot	fell too

Read it across each row: there’s always a green cell. **2022 is the proof** — bonds failed, but the trend sleeve carried the protection. A bonds-only “60/40” had no green that year; Convex Core did.

5 · The convexity, measured (not asserted)

First, the whole 20-year picture — Figure 1's bend, realized. **\$1 invested in 2006 ends up in about the same place** (SPY \$8.50, Convex Core \$7.87) — but watch the path through 2008–09:

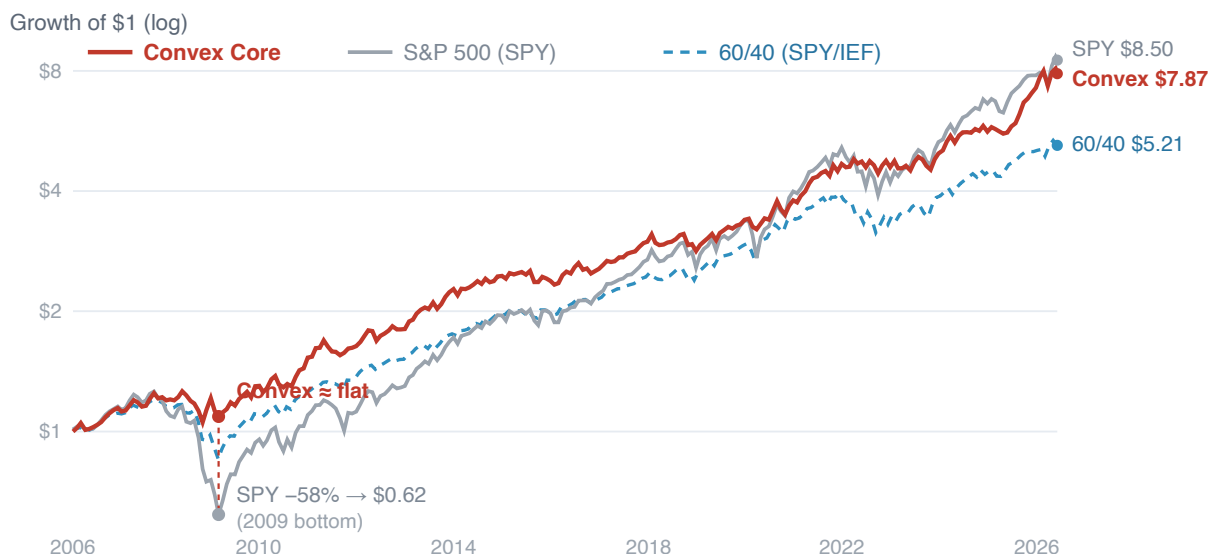
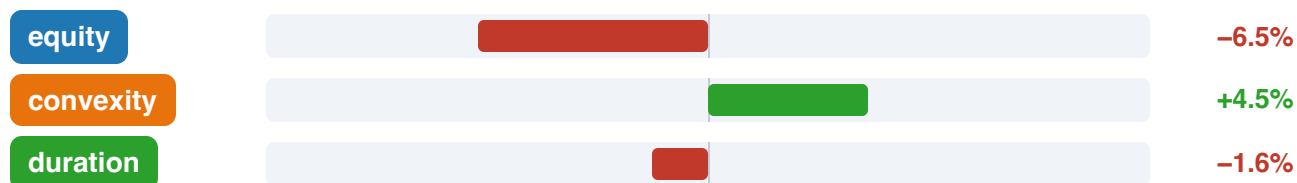


Figure 5. Growth of \$1, 2006→2026, **log scale** (the published backtest, net of modeled costs). Convex Core reaches about the same place as SPY (\$7.87 vs \$8.50) — but at the 2009 bottom the S&P had lost ~58% (down to \$0.62) while Convex Core had barely moved (\$1.09). **That gap is the convexity of Figure 1, lived out:** comparable compounding, a far smoother ride (max drawdown ~-18% vs SPY's ~-55%). The classic **60/40 (SPY/IEF)** balanced portfolio is shown for reference — a smoother ride than SPY, but it gives up a lot of the destination (\$5.21).

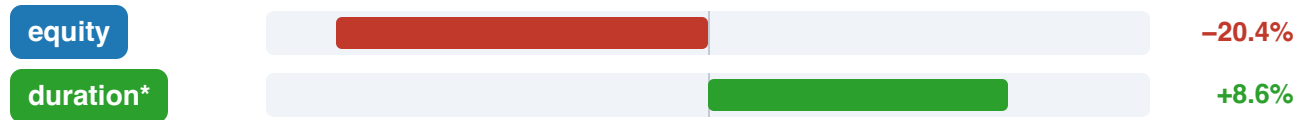
The backtest also decomposes each crisis into how much each sleeve contributed — Figure 1's bend, in numbers:

2022 — stocks *and* bonds fell together



The trend sleeve (+4.5%) did exactly its job in the one year the bond hedge couldn't — it short-sold the very assets that were falling.

2008 — the global financial crisis



Net result: the S&P 500 fell about **-37%**; Convex Core fell about **-1%** — the dot on Figure 1.

*Managed-futures funds didn't exist yet, so the convexity sleeve point-in-time folds into the duration/trend proxy.

6 · The regime engine — reading the market's weather

You've seen two parts of the model move on their own: the **duration sleeve changes flavor** (Section 3) and equity gets **capped in a crisis**. A small, transparent classifier decides which "weather" we're in and drives those changes. It emits **one** primary state plus independent **tags** — all computed point-in-time from local data, no forecasts.

① It picks one state, by precedence

Five mutually-exclusive states are checked **top to bottom**. The first one whose conditions *all* hold wins — so **stress always wins**, and "neutral" is the fallback when the signals disagree.

LIQUIDITY_STRESS — VIX jumping into its top 20% + high-yield cracking vs quality bonds + SPY under its 50/200-day averages → **cap equity at $\leq 0.70\times$ & most-defensive bonds**

RISK_OFF — SPY below its 200-day, 50-day below 200-day (downtrend), VIX in its top 30%, credit deteriorating → **lengthen duration (TLT)**

RECOVERY — recently fell below the averages but is reclaiming them, VIX falling from an elevated level → **begin re-risking**

RISK_ON — SPY above a rising 200-day, positive 63-day return, VIX below its median, breadth >50% above the 200-day → **full equity, neutral bonds**

RISK_NEUTRAL (residual) — mixed signals → **the default; neutral bonds**

Figure 6. The precedence ladder. Checked top-down, first match wins. The order encodes a bias: it's quick to get defensive (stress & risk-off sit at the top) and slower to re-risk (recovery must prove itself).

② What it reads — four families of signals

Every condition is just a fact about *today's* tape, seen through four lenses:

Lens	The actual inputs	What it tells you
 Volatility / fear	VIX 20-day change, its 252-day percentile, VIX vs its own median	Is fear spiking — and how extreme is it versus the last year?
 Credit	HYG vs LQD relative strength (20- & 63-day)	Junk bonds underperforming quality = credit stress building <i>before</i> stocks crack
 Trend / structure	SPY vs its 50- & 200-day moving averages; 63-day return	Is price above or below trend, and rising or falling?
 Breadth	% of stocks above their own 200-day MA	Is the move broad, or a handful of names carrying a thin market?

Graceful degradation. If a signal is temporarily missing, its clause just drops — but a state still needs **≥3 live clauses** to qualify, so a blind input can never silently flip the regime.

③ It refuses to flap — a 3-day confirmation

A new state only takes effect after its conditions hold **three sessions in a row**. A one-day scare can't whipsaw the whole book.

raw daily read →



acted-on state →



◆ day-6 blip ignored (only 1 day)

3 confirming days → flips here ◆

Figure 7. Hysteresis. ■ **RISK_ON** / ■ **RISK_OFF**. The raw read can flicker; the **acted-on** state only changes after 3 straight confirming sessions — so one noisy day costs nothing.

④ What it actually does

The regime never touches the equity/convexity/duration **budget** — it retunes the **duration flavor** and, in stress, caps equity. The same 20% duration sleeve wears a different outfit:

RISK_NEUTRAL / RISK_ON → neutral ladder



RISK_OFF / risk-off rally → lengthen for flight-to-quality



INFLATION_STRESS tag → switch to TIPS (the 2022 fix)



Figure 8. The duration sleeve morphs by regime — nominal Treasuries when bonds are a good hedge, **inflation-protected (VTIP)** when they aren't (2022). **LIQUIDITY_STRESS** additionally hard-caps the equity sleeve at 0.70x on top of the volatility brake.

Five orthogonal **overlay tags** flag specific weather independently of the primary state — any can be on at once:



e.g. **INFLATION_STRESS** fires when energy/materials lead, long bonds are weak, and breakevens are rising — exactly the signal that flips the bond hedge to TIPS.

Honest caveat — the regime layer is a refinement, not the engine. In walk-forward testing, turning it *fully off* moves Sortino only **1.08 → 1.06**. The **volatility brake** (Figure 3) — which reacts in *days* with no classification lag — does most of the de-risking; the regime layer's job is the finer one: pick the right *kind* of bond hedge and add the liquidity-stress cap. It's kept because it's cheap, transparent and robust — not because it carries the result.

7 · So how did the equation come to be?

It's the shortest expression of four beliefs, each backed by published evidence:

1. **Own the market.** Stocks drift up over time, so the core must always participate → a **w_equity** term that never exits.
2. **Drawdowns, not volatility, destroy compounding.** A -50% hole needs +100% back (Figure 2) → spend part of the budget on **crash insurance**, not on chasing a bit more return.
3. **No single hedge is enough** (Section 4) → split the insurance into **two** orthogonal convex sleeves: trend-following (**w_convexity**) and Treasuries (**w_duration**).
4. **Don't overfit.** The weights are frozen a-priori (0.65 / 0.15 / 0.20), not optimized to the past; only the glidepath dial and the mechanical vol-brake move. Because the three must fully invest one dollar, they **sum to 1**, and duration takes the residual.

Put together: **w_equity** gives you the upside line; **w_convexity + w_duration** bend it up in the tail; the brake trims equity when storms hit. The result is the asymmetric, **convex** payoff of Figure 1 — keep most of the upside, lose far less of the downside.

One-sentence summary: Convex Core owns the market with most of its money, then spends the rest on two different kinds of crash insurance that each pay off in a different kind of crisis — so the portfolio's return curve **bends in your favor** exactly when it matters.

8 · An open question — could the equity core be different?

The equity sleeve is **SPY plus a measured tilt overlay** (Section 3). A natural question: should the core be something else — total-market **VTI**, international **VXUS**, small-cap **IWM** — and should it lean by **valuation / CAPE** when the U.S. looks expensive? It's worth being precise here, because the model already has an opinion.

Candidate	What it is	Valuation lean	How Convex Core already reaches it
SPY	S&P 500 — large-cap U.S. (today's base)	The expensive end (elevated CAPE)	It <i>is</i> the core — the “straight line” of Figure 1.
VTI	Total U.S. market (adds mid/small)	≈ SPY (large-caps dominate the cap-weight)	Negligible difference — historically tracks SPY within noise.

VXUS	Total international ex-U.S.	Cheaper (lower CAPE), currency + diversification	The <code>intl_equity</code> tilt sleeve — added <i>when its score earns a slot</i> , not as a fixed bet.
IWM	Russell 2000 small-caps	Cheaper / value-ish, higher volatility	Factor exposure comes through the scored tilts (e.g. MTUM, VLUE) — more surgical than a blanket small-cap bet.

What CAPE is. The Shiller CAPE divides price by the last 10 years of *real* earnings — a valuation gauge. A high CAPE has historically meant *lower* 10-year forward returns. The U.S. (SPY) has sat at an elevated CAPE for years while international, value and small-cap have looked cheaper — which is the whole case for a valuation tilt.

So why is the core still SPY? Two honest reasons, both from this project’s own research:

- ① **We tested CAPE timing — it didn’t earn its keep.** A full CAPE-driven allocation (best config: a rolling-30-year baseline + a trend tilt) reached a Sharpe of about **0.78** — well short of Convex Core’s **1.32**. CAPE tells you about *long-run expected return*, not *when* to act; as a timing signal it’s weak and slow (decade horizon, enormous noise).
- ② **“Cheap” stayed cheap.** “Tilt to the cheapest market” has been a *losing* trade for ~15 years — the U.S. has been the expensive one the whole time and kept winning. And a separate universe-refresh experiment found that swapping in more/younger funds mostly adds **overfitting surface**, **not edge** (verdict: *don’t modernize*).

The model’s answer is already a valuation tilt — just a measured one. Instead of hard-coding VXUS or IWM, the IC-scored overlay leans toward international, value or small-cap *exactly when the data rewards it*, with a guaranteed value slot and a de-dup guard. A **static CAPE/valuation overlay** is a fair candidate for a *pre-registered, gated experiment* — it would have to beat the current tilt net of costs and survive walk-forward before it ships — not a default change. That’s the same parameter-minimal, prove-it-or-publish-the-negative-result discipline behind every other knob in the model. (This applies identically to Convex Prime, which runs the same equity sleeve, levered.)

Full method + significance testing in the Convex Core paper (the  /  links on the model card). © 2026 DNSR Investments, LLC.